

Introduction to Neural Networks

Design of experiment (DOE)

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April 23, 2026©



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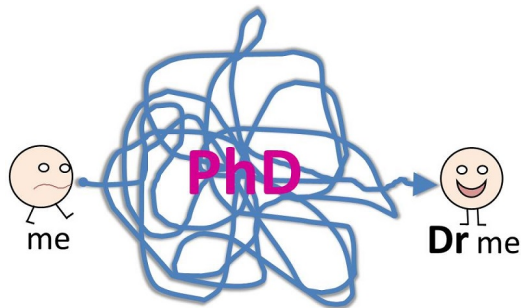
- 1 Introduction
- 2 Design of experiment
- 3 Low level: Hyper-parameter optimization
- 4 Rule of thumb

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The problem

- Training neural networks can be tricky
- This applies to Research and Learning
- Experimentation can lead to nowhere



The solution

- The wrong way
 - Feelings
 - Asking in facebook
 - Asking chatgpt?

The solution

- The wrong way
 - Feelings
 - Asking in facebook
 - Asking chatgpt?
- The good way - decrease the uncertainty
 - Experimental design
 - Hyper-parameter optimization



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Design of experiment

Terminology

Subject

Task and dataset (Classification, detection, etc)

Factor

Controlled independent variable (Hyper-parameter, architecture, ...)

Treatment

Set of factors applied to an experimental unit (Trained architecture)

Experimental unit

Single training (Training with validation)

Experiment

Record of treatments (Trained networks) applied to a set of subjects (tasks).

Design of experiments

Principles

Randomization Process of assigning treatments to experimental units. In DL subjects can receive multiple treatments.

Statistical replication Avoid measurement uncertainty. Replicate experiments that use random variables.

Orthogonality The effects of different factors and interactions can be estimated in the model separately. It usually does not happen in training.

Full-factorial experiment All possible combinations of the levels across all the factors are applied to the experimental units. Usually it is the followed in DL.

Full-factorial experiment

Design space

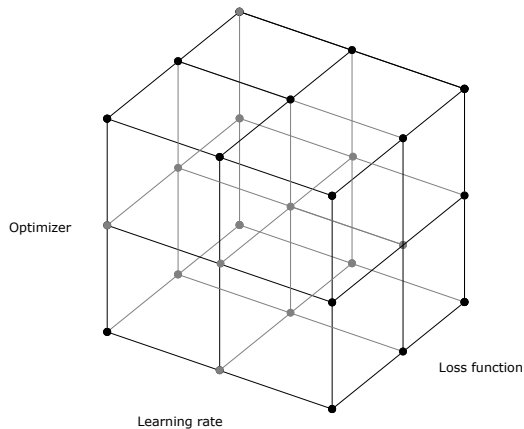


Figure: Full factorial experiment presented on a hypercube in the k-dimensional design space.

Full-factorial experiment

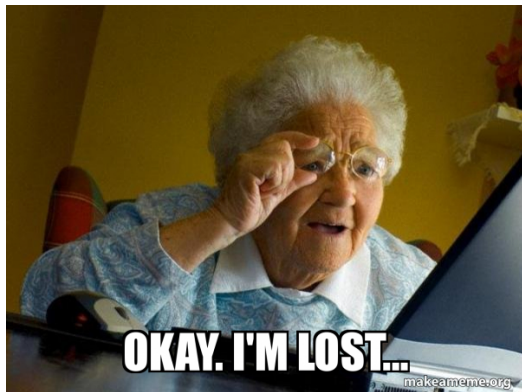
Run	Optimizer	Learning rate	Loss function
1	1	1	1
2	1	1	2
3	1	1	3
4	1	2	1
5	1	2	2

⋮

Table: Example of a full-factorial table.

- Several analysis can be performed
 - Multiple linear regression
 - Analysis of variance
 - etc.
- Such analysis can be overwhelming
- Sometimes our target is only to get "good" accuracy

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Split the experimental design

- "High level": Formal analysis between models
 - ¿Which is the best model?
 - The main target of our research
 - High level threat models
 - Neural network architectures
- "Low level": NN Hyper-parameter optimization
 - How do I get the best performance for a model?
 - Get the best set of hyperparameters
 - It will demonstrate the capabilities of the method

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Hyper-parameter optimization

Successful network training depend on multiple conditions

- Even with fixed subjects and treatments: Data and architectures
- Some hyper-parameters:
 - Learning rate
 - Number of epochs
 - Initialization
 - Optimizer
- What are the best ones?

$$\arg \max_{\theta} \rho(\Phi_{\theta}(X_{val}), Y_{val})$$

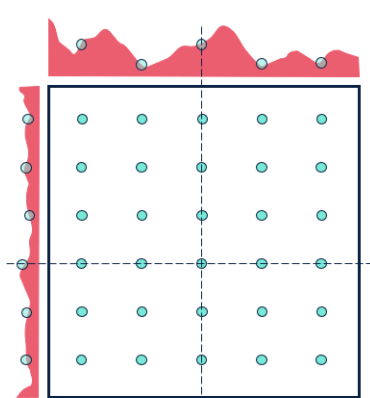
where ρ is a performance metric, such as accuracy or F1 score.

Hyper-parameter optimization

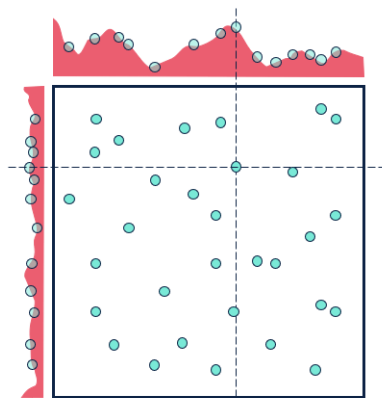
Alternatives

- Closed search space
 - Random search
 - Grid search
- Open search space
 - Optimization algorithms
 - Genetic algorithms
 - Bayesian optimization

Grid search vs random search



Grid Search



Random Search

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Rule of thumb

- 1 Define the problem for the individuals (data, task, metric)
- 2 Define your hypothesis (VGG surpasses GoogleNet in Imagenet)
- 3 Define your design of experiment
 - High level
 - Low level
- 4 Launch experiments

Rule of thumb

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 - High level
 - Low level
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- Kolosova, T., & Berestizhevsky, S. (2020). Supervised Machine Learning: Optimization Framework and Applications with SAS and R. CRC Press.
- SydneyF, Hyperparameter Tuning Black Magic, <https://community.alteryx.com/t5/Data-Science/Hyperparameter-Tuning-Black-Magic/ba-p/449289>